

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ-1

DURATION: 30 MINUTES

WINTER SEMESTER, 2022-2023

FULL MARKS: 15

Math 4543: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

1. Suppose, R is an arbitrary number. Prove that the **Newton-Raphson** formula for finding the square root of R , which is mathematically denoted as $\sqrt{R}$, is –

3

(CO1)

(PO2)

$$x_{i+1} = \frac{1}{2} \left(x_i + \frac{R}{x_i} \right)$$

Solution:

We know, the Newton-Raphson method formula for solving $f(x) = 0$ is,

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Suppose,

$$\begin{aligned} x &= \sqrt{R} \\ \Rightarrow x^2 &= R \\ \Rightarrow x^2 - R &= 0 \end{aligned}$$

We take $f(x) = x^2 - R$ and so $f'(x) = 2x$.

Thus,

$$\begin{aligned} x_{i+1} &= x_i - \frac{x_i^2 - R}{2x_i} \\ &= x_i - \left(\frac{x_i}{2} - \frac{R}{2x_i} \right) \\ &= x_i - \frac{x_i}{2} + \frac{R}{2x_i} \\ &= \frac{x_i}{2} + \frac{R}{2x_i} \\ &= \frac{1}{2} \left(x_i + \frac{R}{x_i} \right) \quad \square \text{ Q.E.D.} \end{aligned}$$

Rubric:

- 1 point for using the Newton-Raphson approximation formula.
- 2 points for correct derivation.

2. a) What are significant digits?

1
(CO1)
(PO1)

Solution:

Significant figures (also known as the significant digits, precision, or resolution) of a number in positional notation are digits in the number that are reliable and necessary to indicate the quantity of something. These are the digits that are known reliably in a reported number.

Rubric:

- 1 point for a correct definition. (Doesn't need to be exactly as written above.)

b) Determine the *minimum* number of significant digits in each of these numerical values —

3
(CO3)
(PO2)

(a) 69.0

(b) 69420

(c) 069420

(d) 0.069420

(e) 69.00420

(f) 6.94200×10^5

Solution:

We need to determine the *minimum* number of significant digits.

- (i) 69.0 has 3 significant digits, since 0's to the right of the decimal point are significant.
- (ii) 69420 has 4 significant digits, since trailing 0's in an integer may or may not be significant.
- (iii) 069420 has 4 significant digits, since leading 0's are not significant and trailing 0's in an integer may or may not be significant.
- (iv) 0.069420 has 5 significant digits, since leading 0's are not significant and trailing 0's in a number with the decimal point are significant.
- (v) 69.00420 has 7 significant digits, since 0's between two significant non-zero digits are significant and trailing 0's in a number with the decimal point are significant.
- (vi) 6.94200×10^5 has 6 significant digits, since all digits of the significand/coefficient of a number's representation using Scientific Notation are significant.

Rubric:

- 0.5 point for each correct answer. (The reasons are not required.)

3. Compare and contrast the following root-finding numerical algorithms— **Bisection** method, **Newton-Raphson** method, **Secant** method, and **False Position** method. 3
(CO3)
(PO2,
PO4)

Solution:

Any 3 valid comparisons will suffice. Some comparisons can be—

Criteria	Bisection	Newton-Raphson	Secant	False Position
Number of initial guesses	2	1	2	2
Condition for initial guesses	Must satisfy $f(x_l)f(x_u) < 0$	Random	Random	Must satisfy $f(x_l)f(x_u) < 0$
Approximation formula	$x_m = \frac{x_l + x_u}{2}$	$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$	$x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$	$x_r = \frac{x_u f(x_l) - x_l f(x_u)}{f(x_l) - f(x_u)}$
Category/Type of method	Bracketed	Open	Open	Bracketed
Guarantee of convergence	Guaranteed	Not guaranteed	Not guaranteed	Guaranteed

Rubric:

- 1 point for each valid comparison.

4. The *Golden Ratio*, often denoted by the Greek letter ϕ (phi), is the ratio of two quantities if their ratio is the same as the ratio of their sum to the larger of the two quantities. Expressed algebraically, for quantities a and b with $a > b > 0$, the ratio $\frac{a+b}{b} = \frac{a}{b} = \phi$ 5
(CO2)
(PO3)
The constant ϕ is equal to the positive root of the quadratic equation $x^2 = x + 1$, which is

$$x_{\text{root}} = \phi = \frac{1 + \sqrt{5}}{2} = 1.618033988749 \dots$$

Use the **Bisection Method** to estimate the value of the golden ratio ϕ . The initial upper and lower guesses are $x_l = 1$ and $x_u = 2$ respectively. Conduct 4 iterations, and find the relative approximate error ($|\epsilon_a|\%$) and the number of significant digits that are at least correct (m), at the end of each iteration. Fill out the following table after performing the necessary calculations.

Iteration	x_l	x_u	x_m	$ \epsilon_a \%$	m	$f(x_m)$
1	1	2		—	—	
2						
3						
4						

Solution:

After performing the series of calculations, we end up with the following table.

Iteration	x_l	x_u	x_m	$ \epsilon_a \%$	m	$f(x_m)$
1	1	2	1.5	—	—	−0.25
2	1.5	2	1.75	14.286	0	0.3125
3	1.5	1.75	1.625	7.692	0	0.015625
4	1.5	1.625	1.5625	4	1	−0.121

Rubric:

- 0.25 point for each correct answer.

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Department of Computer Science and Engineering (CSE)**QUIZ-2****WINTER SEMESTER, 2022-2023****DURATION: 35 MINUTES****FULL MARKS: 15****Math 4543: Numerical Methods****Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).**Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.**STUDENT ID:**

1. The Taylor series for a function
- $f(x)$
- is –

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + f^{(5)}(x)\frac{h^5}{5!} + \dots$$

6
(CO1)
(PO2)

Using the Taylor series (with at least the first 4 terms), derive the Maclaurin series of –

(i) $\ln(1+x)$

(ii) $\ln(1-x)$

Then use these two series to determine the Maclaurin series of $\ln\left(\frac{1+x}{1-x}\right)$.**Solution:**The Maclaurin series of a transcendental function is simply a Taylor series around the point $x = 0$.We know, $\frac{d}{dx}(\ln(x)) = \frac{1}{x}$, and $\frac{d}{dx}(x^n) = nx^{n-1}$.

- (i) Now, for
- $f(x) = \ln(1+x)$
- , let's calculate the derivatives at the point
- $x = 0$
- .

$$f(x) = \ln(1+x), f(0) = \ln(1+0) = \ln(1) = 0; f'(x) = \frac{1}{1+x}, f'(0) = \frac{1}{1+0} = 1;$$

$$f''(x) = -\frac{1}{(1+x)^2}, f''(0) = -\frac{1}{(1+0)^2} = -1; f'''(x) = \frac{2}{(1+x)^3}, f'''(0) = \frac{2}{(1+0)^3} = 2;$$

Using the first 4 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + \dots$$

For $x = 0$,

$$f(0+h) = f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + \dots$$

$$\Rightarrow f(h) = 0 + h - \frac{h^2}{2!} + \frac{2h^3}{3!} + \dots$$

$$\Rightarrow f(x) = 0 + x - \frac{x^2}{2!} + \frac{2x^3}{3!} + \dots$$

$$\Rightarrow \ln(1+x) = 0 + x - \frac{x^2}{2!} + \frac{2x^3}{3!} + \dots$$

$$\Rightarrow \ln(1+x) = x - \frac{x^2}{2!} + \frac{2x^3}{3!} + \dots$$

(ii) **Short version:** Here,

$$\begin{aligned}f(x) &= \ln(1 - x) \\&= \ln(1 + (-x)) \\&= (-x) - \frac{(-x)^2}{2!} + \frac{2(-x)^3}{3!} + \dots \quad [\text{using the Maclaurin series of } \ln(1 + x) \text{ from (i)}] \\&= -x - \frac{x^2}{2!} - \frac{2x^3}{3!} + \dots\end{aligned}$$

Long version: Now, for $f(x) = \ln(1 - x)$, let's calculate the derivatives at the point $x = 0$.

$$f(x) = \ln(1 - x), f(0) = \ln(1 - 0) = \ln(1) = 0; f'(x) = -\frac{1}{1 - x}, f'(0) = -\frac{1}{1 - 0} = -1;$$

$$f''(x) = -\frac{1}{(1 - x)^2}, f''(0) = -\frac{1}{(1 - 0)^2} = -1;$$

$$f'''(x) = -\frac{2}{(1 - x)^3}, f'''(0) = -\frac{2}{(1 - 0)^3} = -2;$$

Using the first 4 terms of the Taylor series,

$$f(x + h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + \dots$$

For $x = 0$,

$$f(0 + h) = f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + \dots$$

$$\Rightarrow f(h) = 0 - h - \frac{h^2}{2!} - \frac{2h^3}{3!} + \dots$$

$$\Rightarrow f(x) = 0 - x - \frac{x^2}{2!} - \frac{2x^3}{3!} + \dots$$

$$\Rightarrow \ln(1 - x) = 0 - x - \frac{x^2}{2!} - \frac{2x^3}{3!} + \dots$$

$$\Rightarrow \ln(1 - x) = -x - \frac{x^2}{2!} - \frac{2x^3}{3!} + \dots$$

We know, $\ln\left(\frac{a}{b}\right) = \ln(a) - \ln(b)$.

So,

$$\begin{aligned}\ln\left(\frac{1 + x}{1 - x}\right) &= \ln(1 + x) - \ln(1 - x) \\&= \left(x - \frac{x^2}{2!} + \frac{2x^3}{3!} + \dots\right) - \left(-x - \frac{x^2}{2!} - \frac{2x^3}{3!} + \dots\right) \\&= 2x + \frac{4x^3}{3!} + \dots\end{aligned}$$

Rubric:

- 2.5 points for the correct series of $\ln(1 + x)$.
- 2.5 points for the correct series of $\ln(1 - x)$.
- 1 point for the correct series of $\ln\left(\frac{1 + x}{1 - x}\right)$.

2. Compare and contrast the following interpolation methods— **Direct** method, **Newton's Divided Difference** method, and **Lagrangian** method. 2
(CO3)
(PO2, PO4)

Solution:

Any 2 valid comparisons will suffice. Some comparisons can be—

Criteria	Direct	Newton's Divided Difference	Lagrangian
Interpolant Polynomial Representation	$f_n(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$	$f_n(x) = b_0 + b_1(x - x_0) + \cdots + b_n(x - x_0)(x - x_1) \cdots (x - x_{n-1})$	$f_n(x) = \sum_{i=0}^n \left(\prod_{j=0, j \neq i}^n \frac{x - x_j}{x_i - x_j} \right) f(x_i)$
Computational Complexity	Depends on the algorithm used to solve the equations. It may be $O(n^3)$, $O(n^3 \log(\ A\ + \ b\))$, or $O((n+1)!)$.	Using memoized dynamic programming, we get a complexity of $O(n^2)$.	The naïve implementation is $O(n^2)$, but it can be optimized further down to $O(n \log^2(n))$.
Property of the terms	Each term has a coefficient a_i and these coefficients are obtained by solving all the equations simultaneously.	Each term has a coefficient b_i and each of these coefficients can be obtained by solving one equation.	The polynomial is a weighted sum of the given functional values $f(x_i)$, and each term is a product of the Lagrangian weight function $L_i(x)$ and the functional value $f(x_i)$.
Vulnerable to the Runge Phenomenon?	Yes	Yes	Yes

Rubric:

- 1 point for each valid comparison.

3. The specific heat capacity C of a substance is defined as the amount of heat that is required to raise the temperature of unit mass of that particular substance by 1 degree. Suppose, you are conducting an experiment to determine how much heat is required to bring some volume of water to its boiling point. The values of specific heat capacity C of water that you have calculated at different temperature values T are shown in the table below –

Temperature, T ($^{\circ}C$)	Specific Heat, C ($Jkg^{-1}^{\circ}C^{-1}$)
22	4181
42	4179
52	4186
82	4199
100	4217

Table 1: Specific heat C of water as a function of temperature T .

Determine the value of the specific heat at $T = 61^{\circ}C$ using the **Direct method** of interpolation and a second order polynomial.

Solution:

For second order polynomial interpolation (also called quadratic interpolation), the equation for the specific heat would be –

$$C(T) = a_0 + a_1T + a_2T^2$$

Here, the order $n = 2$. Now, let's choose the nearest $n + 1 = 2 + 1 = 3$ points that bracket the point $T = 61^{\circ}C$ and evaluate the polynomial. The 3 points are $T_0 = 42$, $T_1 = 52$, and $T_2 = 82$. The respective specific heat values are $C(T_0) = 4179$, $C(T_1) = 4186$, and $C(T_2) = 4199$. Using these values, we obtain the following equations –

$$C(42) = a_0 + a_1 \times 42 + a_2 \times (42)^2 = 4179$$

$$C(52) = a_0 + a_1 \times 52 + a_2 \times (52)^2 = 4186$$

$$C(82) = a_0 + a_1 \times 82 + a_2 \times (82)^2 = 4199$$

Writing these equations in matrix form, we have –

$$\begin{aligned} \begin{bmatrix} 1 & 42 & 1764 \\ 1 & 52 & 2704 \\ 1 & 82 & 6724 \end{bmatrix} \cdot \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix} \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 1 & 42 & 1764 \\ 1 & 52 & 2704 \\ 1 & 82 & 6724 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} \frac{533}{50} & -\frac{287}{25} & \frac{91}{50} \\ -\frac{67}{200} & \frac{31}{75} & -\frac{47}{600} \\ \frac{1}{400} & -\frac{1}{300} & \frac{1}{1200} \end{bmatrix} \cdot \begin{bmatrix} 4179 \\ 4186 \\ 4199 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} 4135 \\ 1.3267 \\ -6.6667 \times 10^{-3} \end{bmatrix} \end{aligned}$$

Hence,

$$C(T) = 4135 + 1.3267T - 6.6667 \times 10^{-3}T^2; \quad 42 \leq T \leq 82$$

At the point $T = 61$, we get,

$$C(61) = 4135 + 1.3267 \times 61 - 6.6667 \times 10^{-3} \times (61)^2 = 4191.2 Jkg^{-1}^{\circ}C^{-1}$$

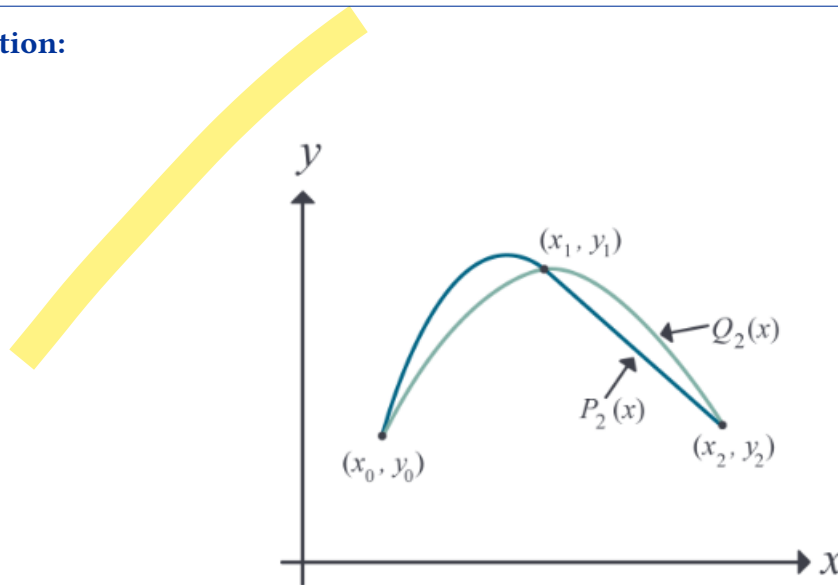
Rubric:

- 1 point for the correct polynomial representation.
- 1 point for choosing the correct nearest points.
- 1 point for the correct coefficient values.
- 1 point for the correct answer.

4. Prove that a polynomial of degree n or less that passes through $n + 1$ data points is unique.

3
(CO1)
(PO2)

Solution:



Proof: Let us use proof by contradiction. If the polynomial is not unique, then at least two polynomials of order n or less pass through the $n + 1$ data points.

Assume two polynomials $P_n(x)$ and $Q_n(x)$ go through $n + 1$ data points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$. Then,

$$R_n(x) = P_n(x) - Q_n(x)$$

Since $P_n(x)$ and $Q_n(x)$ pass through all the $n + 1$ data points, we can say,

$$P_n(x_i) = Q_n(x_i) \quad \forall i \in \{0, \dots, n\}$$

Hence

$$\begin{aligned} R_n(x_i) &= P_n(x_i) - Q_n(x_i) \quad \forall i \in \{0, \dots, n\} \\ \Rightarrow R_n(x_i) &= 0 \end{aligned}$$

The n th order polynomial $R_n(x)$ has $n + 1$ zeros. A polynomial of order n can have $n + 1$ zeros only if it is identical to a zero polynomial, that is,

$$\begin{aligned} R_n(x) &\equiv 0 \\ \Rightarrow P_n(x) &\equiv Q_n(x) \end{aligned}$$

So, $P_n(x)$ and $Q_n(x)$ must be the same unique polynomial, which contradicts our initial assumption. $\square$ Q.E.D.

Rubric:

- 1 point for correctly defining $P_n(x)$, $Q_n(x)$, and $R_n(x)$.
- 1 point for showing $R_n(x)$ is a zero polynomial.
- 1 point for the final conclusion.

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QUIZ-3

DURATION: 30 MINUTES

WINTER SEMESTER, 2022-2023

FULL MARKS: 15

Math 4543: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

1. The zero y -intercept version of the linear regression model for predicting the response for a given set of n data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ is 3 + 2
- $$y = a_1 x$$
- (CO4)

where a_1 is the only parameter of the regression model. (PO1)

- i. Derive the formula for finding the optimal value of a_1 .
- ii. Prove that the value of a_1 obtained using the formula derived in Question-1(i) is unique.

Solution:

- i. The residual error for each given data point is,

$$E_i = y_i - a_1 x_i$$

So the sum of the square of the residual errors is,

$$S_r = \sum_{i=1}^n E_i^2 = \sum_{i=1}^n (y_i - a_1 x_i)^2$$

To find a_1 , we look for the value of a_1 for which S_r is the absolute minimum. At first, let's do the first derivative test.

$$\begin{aligned} \frac{dS_r}{da_1} &= 2 \sum_{i=1}^n (y_i - a_1 x_i)(-x_i) = 0 \\ \Rightarrow -2 \sum_{i=1}^n y_i x_i + 2 \sum_{i=1}^n a_1 x_i^2 &= 0 \Rightarrow -2 \sum_{i=1}^n y_i x_i + 2a_1 \sum_{i=1}^n x_i^2 = 0 \\ \Rightarrow a_1 &= \frac{\sum_{i=1}^n y_i x_i}{\sum_{i=1}^n x_i^2} \end{aligned}$$

Rubric:

- 0.5 point for the correct equation of E_i .
- 1 point for the correct equation of S_r .
- 1.5 points for the correct derivation of a_1 .

- ii. To prove the uniqueness of a_1 , we need to prove that its value corresponds to the absolute minimum of S_r . For this, let's invoke the second derivative test.

$$\begin{aligned}\frac{d^2 S_r}{da_1^2} &= \frac{d}{da_1} \left(\frac{dS_r}{da_1} \right) \\ &= \frac{d}{da_1} \left(2 \sum_{i=1}^n (y_i - a_1 x_i)(-x_i) \right) \\ &= \frac{d}{da_1} \left(\sum_{i=1}^n (-2x_i y_i + 2a_1 x_i^2) \right) \\ &= \sum_{i=1}^n 2x_i^2 > 0\end{aligned}$$

by pragmatically assuming that all the x_i values are not zero at the same time. Hence, S_r yields the absolute minimum value at that value of a_1 . So the value of a_1 that is obtained using the formula in Question-1(i) is unique. $\square$ Q.E.D.

Rubric:

- 1.5 points for the correct second derivative test.
- 0.5 point for the correct conclusion.

2. What are the types of processes that can be represented using the Exponential model of nonlinear regression? Briefly explain with suitable real-world examples.

3
(CO3)
(PO1)

Solution:

The exponential model can be used to represent –

- **Exponentially increasing processes:** Population growth of a species in their natural habitat is governed by the equation $P_t = P_0 e^{kt}$.
- **Exponentially decaying processes:** Radiation intensity of a radioactive Tc-99 isotope generally follows the equation $\gamma = Ae^{-\lambda t}$.

Rubric:

- 1 point for each correct process type.
- 0.5 point for each correct real-world example.

3. Suppose, you are monitoring the progress of a homogeneous chemical reaction in order to calculate the rate constant k and the order of the reaction n . The rate law expression for the reaction can be represented with the Power model

$$-r = kC^n$$

where r is the reaction rate and C is the concentration of the reactant. The values of reaction rate $-r$ and the respective concentration C of the reactant that you have calculated over the course of time are shown in the table below –

Concentration, C ($gmolL^{-1}$)	Reaction Rate, $-r$ ($gmolL^{-1}s^{-1}$)
4	0.398
2.25	0.298
1.45	0.238
1	0.198
0.65	0.158
0.25	0.098
0.006	0.048

Table 1: Chemical kinetics of the observed homogeneous reaction.

Determine the value of the rate constant k and the order of the reaction n using the data provided in Table-1. You are allowed to apply data transformation.

Solution:

Let's linearize the data at first. Taking the natural logarithm of both sides of the given rate law expression, we get,

$$\ln(-r) = \ln(k) + n \ln(C)$$

Let, $z = \ln(-r)$, $w = \ln(C)$, and $a_0 = \ln(k)$; implying that $k = e^{a_0}$ and $n = a_1$. The straight line that we end up with is

$$z = a_0 + a_1 w$$

Using the calculator, we obtain the following values for the $\hat{n} = 7$ transformed data points.

$$\sum_{i=1}^{\hat{n}} w_i = -4.3643, \sum_{i=1}^{\hat{n}} z_i = -12.391, \sum_{i=1}^{\hat{n}} w_i z_i = 16.758, \sum_{i=1}^{\hat{n}} w_i^2 = 30.998$$

Now, let's apply the formulae for determining the corresponding linear regression parameters a_0 and a_1 .

$$a_1 = \frac{\hat{n} \sum_{i=1}^{\hat{n}} w_i z_i - \sum_{i=1}^{\hat{n}} w_i \sum_{i=1}^{\hat{n}} z_i}{\hat{n} \sum_{i=1}^{\hat{n}} w_i^2 - \left(\sum_{i=1}^{\hat{n}} w_i \right)^2} = 0.31943, a_0 = \frac{\sum_{i=1}^{\hat{n}} z_i}{\hat{n}} - a_1 \frac{\sum_{i=1}^{\hat{n}} w_i}{\hat{n}} = -1.5711$$

Hence, we obtain

$$k = e^{-1.5711}$$

$$= 0.20782$$

$$n = 0.31941$$

Plugging these values into the given Power model equation, we get,

$$-r = 0.20782 \times C^{0.31941}$$

Rubric:

- 1 point for the correct linear representation.
- 1 point for the correct a_1 formula.
- 0.5 point for the correct a_1 value.
- 1 point for the correct a_0 formula.
- 0.5 point for the correct a_0 value.
- 0.5 point for the correct k value.
- 0.5 point for the correct n value.

4. Geometrically derive the integral approximation formula of the Single-segment Trapezoidal rule.

2
(CO4)
(PO1)

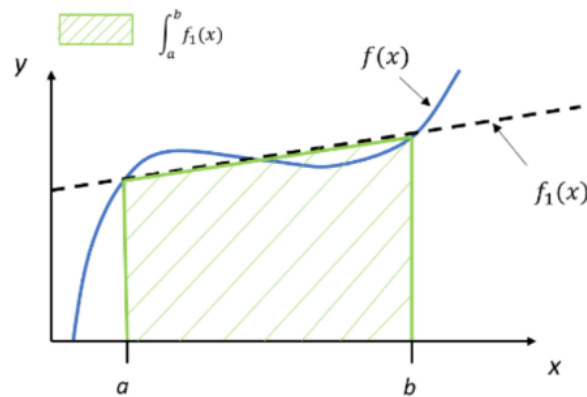
Solution:

Figure 1: Approximating the function by a first-order polynomial to derive the trapezoidal rule.

We'll approximate the area under the curve $f(x)$ using the area under the straight line $f_1(x)$ within the given interval $[a, b]$, which is essentially a trapezoid (shaded with green color). The integral is –

$$\begin{aligned}
 I = \int_a^b f(x) &\approx \int_a^b f_1(x) = \text{Area of the trapezoid} \\
 &= \frac{1}{2} \times (\text{Sum of parallel sides}) \times (\text{Distance between the parallel sides}) \\
 &= \frac{1}{2} [f(b) + f(a)] \times (b - a) \\
 &= (b - a) \left[\frac{f(a) + f(b)}{2} \right]
 \end{aligned}$$

which is the desired integral approximation formula for the single-segment trapezoidal rule.

Rubric:

- 2 points for a correct geometrical derivation.

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QUIZ-4

DURATION: 30 MINUTES

WINTER SEMESTER, 2022-2023

FULL MARKS: 15

Math 4543: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

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- | | |
|--|-------------------------|
| 1. State the fundamental theorems of calculus. Applying these theorems, explain how Euler's method is used to approximate the value of a definite integral. | 3 + 2
(CO4)
(PO1) |
| 2. Compare and contrast the following numerical algorithms for solving 1st order ODEs – Euler's method, Runge-Kutta 2nd order method, and Runge-Kutta 4th order method. | 2
(CO3)
(PO1) |
| 3. Determine the approximate value of
$I = \int_8^{30} \left[2000 \ln \left(\frac{140000}{140000 - 2100t} \right) - 9.8t \right] dt$ using Simpson's 1/3 rule (with $n_1 = 6$) and Simpson's 3/8 rule (with $n_2 = 6$). | 6
(CO2)
(PO1) |
| 4. An internet meme is defined as a humorous or culturally relevant image, video, or text that spreads rapidly online, often evolving as it is shared and adapted by users. Draw a rough sketch of a math meme related to this course. | 2
(CO69)
(PO420) |
- Warning:** Cringe and lame memes will result in negative marking.

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

WINTER SEMESTER, 2022-2023
FULL MARKS: 75

Math 4543: Numerical Methods

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

1. a) i. Define numerical methods. Briefly explain why we need them.

2
(CO2)
(PO1)

Solution:

Numerical methods are techniques to approximate mathematical processes.

The application of numerical methods is an important step in solving engineering problems. We need to resort to finding approximations because there are some problems that cannot be solved analytically or have intractable analytical solutions.

Rubric:

- 1 point for a correct definition.
- 1 point for a valid explanation.

- ii. How do numerical approaches differ from analytical approaches to solving a problem?

2
(CO2)
(PO1)

Solution:

Any 2 valid differences will suffice. Some differences can be written as follows —

Criteria	Numerical Methods	Analytical Methods
Type of yielded solution	Approximate solution	Exact solution
Working Procedure	Keep iterating until the obtained solution is a “good enough” approximation.	Perform symbolic manipulations to arrive at an equation that yields the exact solution.
Example	Solving $ax^2 + bx + c = 0$ using the quadratic equation $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$	Solving $ax^2 + bx + c = 0$ using any root-finding algorithm e.g. the Bisection method.

Rubric:

- 1 point for each valid difference.

- b) i. Suppose, R is an arbitrary number. Prove that the *Newton-Raphson* formula for finding the inverse of the square root of R , which is mathematically denoted as $\frac{1}{\sqrt{R}}$, is as shown in Equation-1. 5
(CO2)
(PO1)

$$x_{i+1} = \frac{1}{2} \left(x_i + \frac{1}{Rx_i} \right) \quad (1)$$

Solution:

We know, the Newton-Raphson method formula for solving $f(x) = 0$ is,

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Suppose,

$$\begin{aligned} x &= \frac{1}{\sqrt{R}} \\ \Rightarrow x^2 &= \frac{1}{R} \\ \Rightarrow x^2 - \frac{1}{R} &= 0 \end{aligned}$$

We take $f(x) = x^2 - \frac{1}{R}$ and so $f'(x) = 2x$.

Thus,

$$\begin{aligned} x_{i+1} &= x_i - \frac{x_i^2 - \frac{1}{R}}{2x_i} \\ &= x_i - \left(\frac{Rx_i^2 - 1}{2Rx_i} \right) \\ &= x_i - \left(\frac{x_i}{2} - \frac{1}{2Rx_i} \right) \\ &= x_i - \frac{x_i}{2} + \frac{1}{2Rx_i} \\ &= \frac{x_i}{2} + \frac{1}{2Rx_i} \\ &= \frac{1}{2} \left(x_i + \frac{1}{Rx_i} \right) \quad \square \text{ Q.E.D.} \end{aligned}$$

Rubric:

- 1 point for using the Newton-Raphson approximation formula.
- 1 point for defining $f(x)$ and $f'(x)$ properly.
- 3 points for correct derivation.

- ii. What is the advantage of formulating $\frac{1}{\sqrt{R}}$ like Equation-1 using the *Newton-Raphson* method? 2
(CO3)
(PO1)

Solution:

The instructions for the division operator (`fdiv`) and the square root operator (`fsqrt`) comparatively require a lot more clock cycles. The Newton-Raphson formulation of $\frac{1}{\sqrt{R}}$ can be seen as an efficient substitute for these costly mathematical operations.

Rubric:

- 1 point for mentioning that division is costly.
- 1 point for mentioning that square root is costly.

- c) The *Silver Ratio*, often denoted by the Greek letter δ (delta), is the ratio of two quantities a and b with $a > b > 0$, if the ratio $\frac{2a+b}{a} = \frac{a}{b}$. 14
(CO2)
(PO1)
- The constant δ is equal to the positive root of the quadratic equation $x^2 = 2x + 1$, which is
- $$x_{root} = \delta = 1 + \sqrt{2} = 2.41421356237 \dots$$

Apply the *Regula Falsi* method, also known as the *False Position* method, to estimate the value of the silver ratio δ . The initial upper and lower guesses are $x_l = 2$ and $x_u = 3$ respectively. Demonstrating the step-by-step mathematical procedure, conduct 4 iterations, and find the relative approximate error ($|\epsilon_a|\%$) and the number of significant digits that are at least correct (m) at the end of each iteration. Draw Table-1 on your answer script and fill it out after performing the necessary calculations.

Table 1: The relevant values obtained in the answer of Question-1(c).

Iteration	x_l	x_u	x_r	$ \epsilon_a \%$	m	$f(x_r)$
1	2	3		—	—	
2						
3						
4						

Solution:

After performing the series of calculations, we end up with the following table.

Iteration	x_l	x_u	x_r	$ \epsilon_a \%$	m	$f(x_r)$
1	2	3	2.333	—	—	−0.222
2	2.333	3	2.4	2.78	1	−0.04
3	2.4	3	2.4118	0.49	2	−0.0069
4	2.4118	3	2.4138	0.08	2	−0.0002

Rubric:

- 0.5 point for each correct answer.
- 1 point for demonstrating the step-by-step procedure in each iteration.

2. a) Define *Truncation Error*. What are the different avenues through which *Truncation Error* can be introduced? Explain with proper examples.

5
(CO3)
(PO1)

Solution:

Truncation Error is the error caused by approximating the mathematical process or the modeling technique. Some of the ways it can be introduced are —

1. **Chopping off terms in a series:** If we consider only the first 3 terms of the Maclaurin series of e^x , we get $e^x \approx \left(1 + x + \frac{x^2}{2!}\right)$. And so the truncation error is,

$$\text{Truncation Error} = \text{Exact Value} - \text{Approximate Value}$$

$$= e^x - \left(1 + x + \frac{x^2}{2!}\right)$$

2. **Using a finite step size instead of an infinitesimal value:** To find the derivative of a function $f(x)$ using a finite step size h instead of an infinitesimally small h , the truncation error is,

$$\text{Truncation Error} = \text{Exact Value} - \text{Approximate Value}$$

$$= \left(\lim_{h \Rightarrow 0} \frac{f(x+h) - f(x)}{h}\right) - \left(\frac{f(x+h) - f(x)}{h}\right)$$

3. **Approximating area under curve with rectangles:** To find the integral value of a function $f(x)$ within given bounds a and b using n rectangles of width Δx_i and height $f(x_i)$, the truncation error is,

$$\text{Truncation Error} = \text{Exact Value} - \text{Approximate Value}$$

$$= \int_a^b f(x)dx - \sum_{i=0}^n f(x_i) \cdot \Delta x_i$$

Rubric:

- 1 point for a correct definition.
- 2 points for each valid example. Any 2 valid examples will suffice.

b) The Taylor series for a function $f(x)$ is —

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + f^{(5)}(x)\frac{h^5}{5!} + \dots$$

Using the Taylor series (with at least the first 5 terms), derive the Maclaurin series of

$$f(x) = \sec(x)$$

Solution:

The Maclaurin series of a transcendental function is simply a Taylor series around the point $x = 0$.

We know, $\frac{d}{dx}(\sec(x)) = \sec(x)\tan(x)$, and $\frac{d}{dx}(u(x)v(x)) = u(x)\frac{d}{dx}(v(x)) + v(x)\frac{d}{dx}(u(x))$. Now, for $f(x) = \sec(x)$, let's calculate the derivatives at the point $x = 0$.

$$f(x) = \sec(x), \quad f(0) = 1$$

$$f'(x) = \sec(x)\tan(x), \quad f'(0) = 0$$

$$f''(x) = \sec^3(x) + \sec(x)\tan^2(x), \quad f''(0) = 1$$

$$f'''(x) = 5\sec^3(x)\tan(x) + \sec(x)\tan^3(x), \quad f'''(0) = 0$$

$$f^{(4)}(x) = 5\sec^5(x) + 18\sec^3(x)\tan^2(x) + \sec(x)\tan^4(x), \quad f^{(4)}(0) = 5$$

Using the first 5 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + \dots$$

For $x = 0$,

$$f(0+h) = f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + f^{(4)}(0)\frac{h^4}{4!} + \dots$$

$$\Rightarrow f(h) = 1 + 0 + \frac{h^2}{2!} + 0 + \frac{5h^4}{4!} + \dots$$

$$\Rightarrow f(x) = 1 + \frac{x^2}{2!} + \frac{5x^4}{4!} + \dots$$

$$\Rightarrow \sec(x) = 1 + \frac{x^2}{2!} + \frac{5x^4}{4!} + \dots$$

$$\Rightarrow \sec(x) = 1 + \frac{x^2}{2} + \frac{5x^4}{24} + \dots$$

Rubric:

- 1 point for defining $f(x)$ and finding each i -th order derivative $f^i(x)$.
- 1 point for correct values of $f(0)$ and each $f^i(0)$.
- 5 points for deriving the correct series.

c) The Maclaurin series of the exponential function e^x is —

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots$$

i. Why is the exponential function e^x categorized as a *transcendental function*?

1

Solution:

The exponential function e^x cannot be expressed using a finite sequence of terms or algebraic operations which is why it falls under the umbrella of transcendental functions.

Rubric:

- 1 point for the correct reason.

(CO4)

(PO1)

ii. Using the *Remainder Theorem*, establish the bounds of the truncation error in the representation of $e^{1.69}$ if only the first 5 terms of the series are used.

4

(CO2)

(PO1)

Solution:

If we are using 5 terms of the Taylor series, the remainder is given by ($x = 0$, $h = 1.69$, $n = 4$)

$$R_4(0 + 1.69) = \frac{1.69^{(4+1)}}{(4+1)!} f^{(4+1)}(c) = \frac{1.69^5}{5!} f^5(c) = \frac{1.69^5 e^c}{120}$$

Since,

$$\begin{aligned} x &< c < x + h \\ 0 &< c < 0 + 1.69 \\ 0 &< c < 1.69 \end{aligned}$$

The error bound is between,

$$\begin{aligned} \frac{1.69^5 e^0}{120} &< R_4(1.69) < \frac{1.69^5 e^{1.69}}{120} \\ 0.1148 &< R_4(1.69) < 0.6226 \end{aligned}$$

Rubric:

- 1 point for finding $R_4(1.69)$ in terms of c .
- 1.5 points for correct lower-bound.
- 1.5 points for correct upper-bound.

3. a) Differentiate between *Interpolation* and *Extrapolation*. With suitable real-world examples, briefly write when each of the methods is applicable.

4
(CO3)
(PO1)

Solution:

Any 2 valid conceptual differences will suffice. There should be at least 1 valid real-world example that is used to explain the applicability. Some differences are —

Criteria	Interpolation	Extrapolation
Position of newly estimated points	Within the distribution of the known data points.	Outside the distribution of the known data points.
Likelihood of proper estimation	Has a greater likelihood of obtaining a valid estimate, since the pattern of the data points on either side of the newly estimated point is known.	Has a lesser likelihood of obtaining a valid estimate, since it assumes that the observed trend continues for values of x outside the range of the given data points.
Applicability	• Computer Graphics – For constructing B-splines, smooth surfaces, etc. • Carpentry – For drawing curves on wooden surfaces using different apparatus e.g. the Spline tool. • Digital Image Processing – For increasing the resolution of images. • Electronics – For plotting transfer functions or calibration curves of electronic components e.g. Thermistors.	

Rubric:

- 1 point for each valid conceptual difference.
- 2 points for a valid real-world application.

- b) Suppose, there is an unknown function $f(x)$ and you are given 3 points $(x_0, f(x_0))$, $(x_1, f(x_1))$, and $(x_2, f(x_2))$ through which $f(x)$ passes. Now, in order to approximate $f(x)$, you decide to use the *Newton's Divided Difference* method of interpolation and obtain a *quadratic interpolant* $f_2(x)$. Derive this 2nd order Newton's Divided Difference polynomial $f_2(x)$. 8
(CO4)
(PO1)

Solution:



Just [click here](#).

Rubric:

- 1 point for writing the quadratic NDD interpolant $f_2(x)$ in the proper form.
- 1 point for correct derivation of b_0 .
- 1 point for correct derivation of b_1 .
- 4 points for correct derivation of b_2 .
- 1 point for writing the correct expanded form of the quadratic NDD interpolant $f_2(x)$.

- c) i. Briefly explain the necessity of *Spline Interpolation*. 2
(CO3)
(PO1)

Solution:

Other interpolation methods become susceptible to the Runge phenomenon when the order n of the interpolant is gradually increased (due to the manifestation of an unhinged oscillatory behavior at either end of the interpolant). We need Spline interpolation since it is immune to this Runge phenomenon.

Rubric:

- 2 points for mentioning its immunity to the Runge phenomenon.

- ii. Suppose you want to perform Spline interpolation on $n + 1$ data points $(x_0, f(x_0))$, $(x_1, f(x_1))$, $\dots$, and $(x_n, f(x_n))$ using *Cubic splines* of the form $f(x) = ax^3 + bx^2 + cx + d$. How many unknowns and how many equations would you have to deal with? Mention how you would obtain those equations and derive them. 5
(CO4)
(PO1)

Solution:

Each cubic spline has 4 unknowns. So, for n such splines, we have a total of $4n$ unknowns. To obtain these $4n$ unknowns, we need $4n$ equations.

Here, we have $n + 1$ given data points $(x_0, f(x_0))$, $(x_1, f(x_1))$, $\dots$, and $(x_n, f(x_n))$. The n cubic splines that go through consecutive points and form the interpolant $f(x)$ are,

$$f(x) = \begin{cases} a_1x^3 + b_1x^2 + c_1x + d_1; & \text{for } x_0 \leq x \leq x_1 \\ a_2x^3 + b_2x^2 + c_2x + d_2; & \text{for } x_1 \leq x \leq x_2 \\ \vdots & \vdots \\ a_nx^3 + b_nx^2 + c_nx + d_n; & \text{for } x_{n-1} \leq x \leq x_n \end{cases}$$

Now, let's derive the $4n$ equations. Since the cubic splines pass through consecutive points, we get the following $2n$ equations.

$$\begin{aligned}
 a_1x_0^3 + b_1x_0^2 + c_1x_0 + d_1 &= f(x_0) \\
 a_1x_1^3 + b_1x_1^2 + c_1x_1 + d_1 &= f(x_1) \\
 \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\
 a_1x_{i-1}^3 + b_1x_{i-1}^2 + c_1x_{i-1} + d_1 &= f(x_{i-1}) \\
 a_1x_i^3 + b_1x_i^2 + c_1x_i + d_1 &= f(x_i) \\
 \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\
 a_1x_{n-1}^3 + b_1x_{n-1}^2 + c_1x_{n-1} + d_1 &= f(x_{n-1}) \\
 a_1x_n^3 + b_1x_n^2 + c_1x_n + d_1 &= f(x_n)
 \end{aligned}$$

Now, find out the 1st order derivatives of the consecutive cubic splines at the $n - 1$ interior points, since they are continuous at those points. We obtain another $n - 1$ equations in this way.

$$\begin{aligned}
 3a_1x_1^2 + 2b_1x_1 + c_1 &= 3a_2x_1^2 + 2b_2x_1 + c_2 \\
 \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\
 3a_{n-1}x_{n-1}^2 + 2b_{n-1}x_{n-1} + c_{n-1} &= 3a_nx_{n-1}^2 + 2b_nx_{n-1} + c_n
 \end{aligned}$$

Now, find out the 2nd order derivatives of the consecutive cubic splines at the $n - 1$ interior points, since they are continuous at those points. We obtain yet another $n - 1$ equations in this way.

$$\begin{aligned}
 6a_1x_1 + 2b_1 &= 6a_2x_1 + 2b_2 \\
 \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\
 6a_{n-1}x_{n-1} + 2b_{n-1} &= 6a_nx_{n-1} + 2b_n
 \end{aligned}$$

Now, we require 2 more equations. For these, we consider $a_1 = 0$ and $a_{n-1} = 0$. So, in aggregate, we have $2n + (n - 1) + (n - 1) + 2 = 4n$ equations for the $4n$ unknowns.

Rubric:

- 1 point for the correct number of unknowns and equations.
- 1 point for the initial $2n$ equations.
- 1 point for the $n - 1$ equations using 1st order derivatives.
- 1 point for the $n - 1$ equations using 2nd order derivatives.
- 1 point for the last 2 equations.

- d) A thermistor, also known as thermal resistor, is a semiconductor type of resistor whose resistance is strongly dependent on temperature. To measure temperature using a thermistor, the manufacturers provide users with a Temperature (T) vs. Resistance (R) calibration curve. If we measure the resistance of the thermistor using an Ohmmeter, then we can refer to the aforementioned curve and determine the corresponding temperature value. Figure-1 and Table-2 portray multiple recorded observations involving a thermistor.

Table 2: Temperature T of a thermistor as a function of its resistance R for Question-3(d).

Resistance, R (Ω)	Temperature, T ($^{\circ}C$)
1101	25.113
911.3	30.131
636	40.12
451.1	50.128

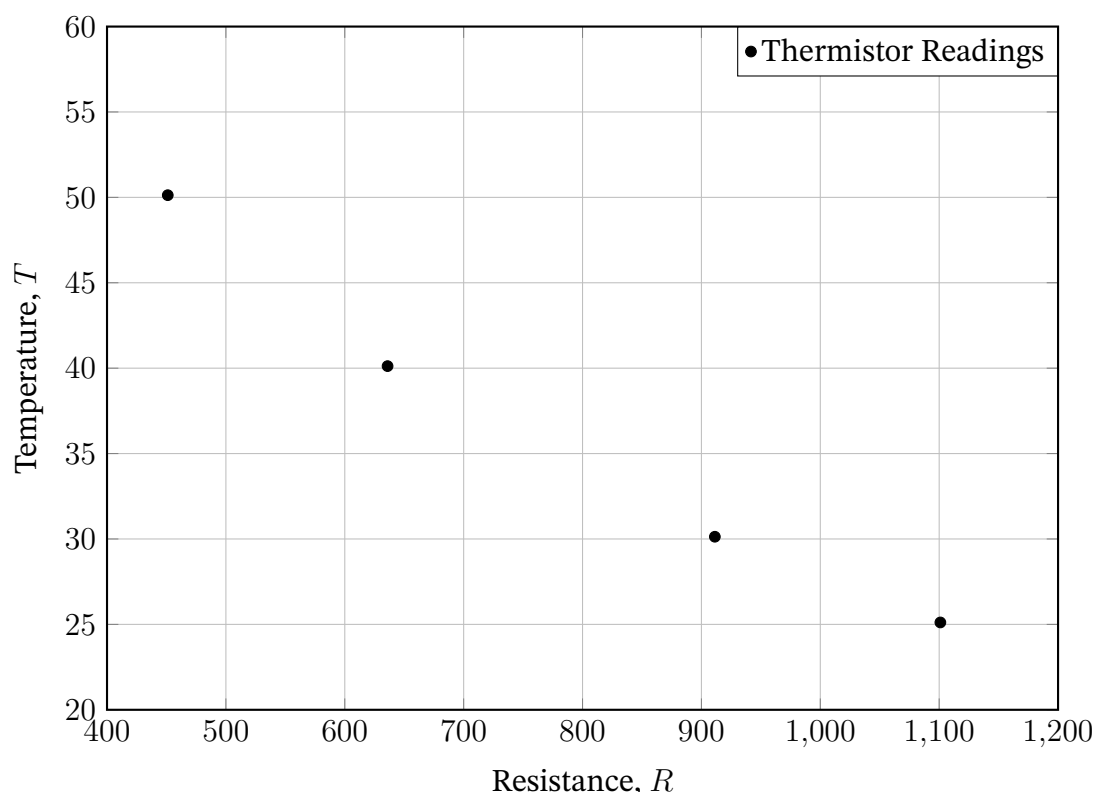


Figure 1: Temperature T vs. Resistance R for Question-3(d).

Determine the value of the temperature T when the thermistor resistance is $R = 754.8\Omega$ using the *Lagrangian method* of interpolation and a second order polynomial.

Solution:

The 2nd order Lagrangian interpolant is —

$$\begin{aligned} T(R) &= \sum_{i=0}^2 L_i(R)T(R_i) \\ &= L_0(R)T(R_0) + L_1(R)T(R_1) + L_2(R)T(R_2) \end{aligned}$$

Given, $R = 754.8\Omega$, and the 3 nearest data points that bracket it are, $R_0 = 911.3$, $R_1 = 636$, and $R_2 = 451.1$. The corresponding T values are $T(R_0) = 30.131$, $T(R_1) = 40.12$, and $T(R_2) = 50.128$ respectively.

Now, let's calculate the Lagrangian weights.

$$L_0(R) = \prod_{\substack{j=0 \\ j \neq 0}}^2 \frac{R - R_j}{R_0 - R_j} = \left(\frac{754.8 - 636}{911.3 - 636} \right) \times \left(\frac{754.8 - 451.1}{911.3 - 451.1} \right) = 0.284$$

$$L_1(R) = \prod_{\substack{j=0 \\ j \neq 1}}^2 \frac{R - R_j}{R_0 - R_j} = \left(\frac{754.8 - 911.3}{636 - 911.3} \right) \times \left(\frac{754.8 - 451.1}{636 - 451.1} \right) = 0.933$$

$$L_2(R) = \prod_{\substack{j=0 \\ j \neq 2}}^2 \frac{R - R_j}{R_0 - R_j} = \left(\frac{754.8 - 911.3}{451.1 - 911.3} \right) \times \left(\frac{754.8 - 636}{451.1 - 636} \right) = -0.218$$

Now that we have the weights, we can obtain the value of $T(754.8)$.

$$\begin{aligned} T(754.8) &= 0.284 \times T(911.3) + 0.933 \times T(636) + (-0.218) \times T(451.1) \\ &= 0.284 \times 30.131 + 0.933 \times 40.12 + (-0.218) \times 50.128 \\ &\approx 35.088^\circ\text{C} \end{aligned}$$

Rubric:

- 1 point for the correct polynomial representation.
- 1 point for choosing the correct nearest points.
- 1 point for each Lagrangian weight $L_i(R)$ value.
- 1 point for the correct answer.

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

WINTER SEMESTER, 2022-2023
FULL MARKS: 150

Math 4543: Numerical Methods

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 6 (six)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

-
1. a) What are the steps of problem-solving in engineering? With a suitable example, briefly explain how numerical methods are used to solve an engineering problem. 2 + 1
(CO2)
(PO1)

Solution:

The steps of problem-solving in engineering are —

1. Describing or formulating the problem.
2. Using a mathematical model to represent the problem.
3. Solving the mathematical model.
4. Using the solution of the mathematical model to solve the original problem.

e.g. Let's consider a kinematics problem. Given the initial velocity u and the height h of a falling body, we can calculate the time t it needs to reach the ground by finding the positive root of the equation $ut + \frac{1}{2}gt^2 - h = 0$ using a root-finding algorithm.

- b) i. Suppose, R and k are arbitrary numbers such that $R > 0$ and $k \neq 0$. Prove that the Newton-Raphson formula for finding the k th root of R , which is mathematically denoted as $\sqrt[k]{R}$, is as shown in Equation-1. 5
(CO2)
(PO1)

$$x_{i+1} = \frac{1}{k} \left[(k-1)x_i + \frac{R}{x_i^{k-1}} \right] \quad (1)$$

Solution:

We know, the Newton-Raphson method formula for solving $f(x) = 0$ is,

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Suppose,

$$\begin{aligned} x &= \sqrt[k]{R} \\ \Rightarrow x^k &= R \\ \Rightarrow x^k - R &= 0 \end{aligned}$$

We take $f(x) = x^k - R$ and so $f'(x) = kx^{k-1}$.

Thus,

$$\begin{aligned} x_{i+1} &= x_i - \frac{x_i^k - R}{kx_i^{k-1}} \\ &= \frac{kx_i^k - x_i^k + R}{kx_i^{k-1}} \\ &= \frac{(k-1)x_i^k + R}{kx_i^{k-1}} \\ &= \frac{1}{k} \left[\frac{(k-1)x_i^k + R}{x_i^{k-1}} \right] \\ &= \frac{1}{k} \left[(k-1)x_i + \frac{R}{x_i^{k-1}} \right] \quad \square \text{ Q.E.D.} \end{aligned}$$

- ii. Draw a rough sketch of the graph of two functions $f_1(x)$ and $f_2(x)$ given that —
- For $f_1(x)$, False Position method performs better than the Bisection method.
 - For $f_2(x)$, Bisection method performs better than the False Position method.

2 + 2

(CO3)

(PO2)

Here, “performs better” means that it takes a lesser number of iterations to converge to the root. You have the liberty to choose the initial guesses $\langle x_l, x_u \rangle$ as well.

Solution:

$f_1(x)$ may look something like,

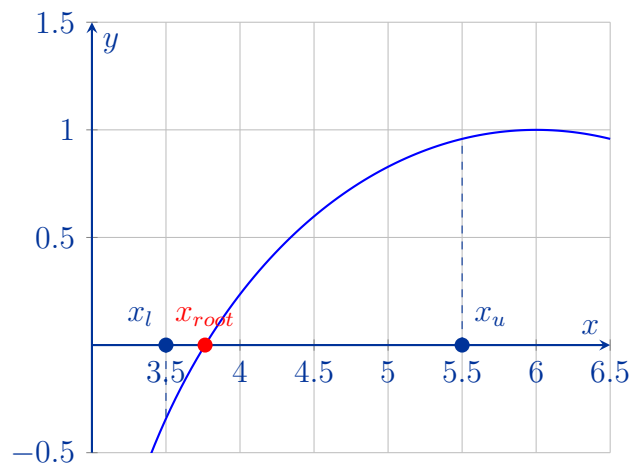


Figure 1: Graph of $f_1(x)$.

$f_2(x)$ may look something like,

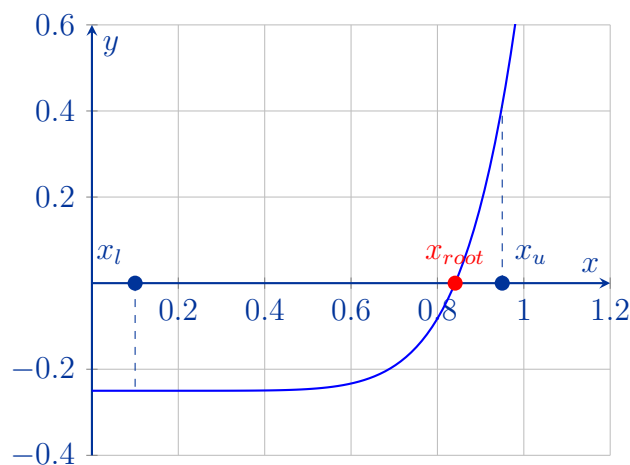


Figure 2: Graph of $f_2(x)$.

iii. Derive the formula for the Secant method geometrically.

4

(CO4)

(PO1)

Solution:

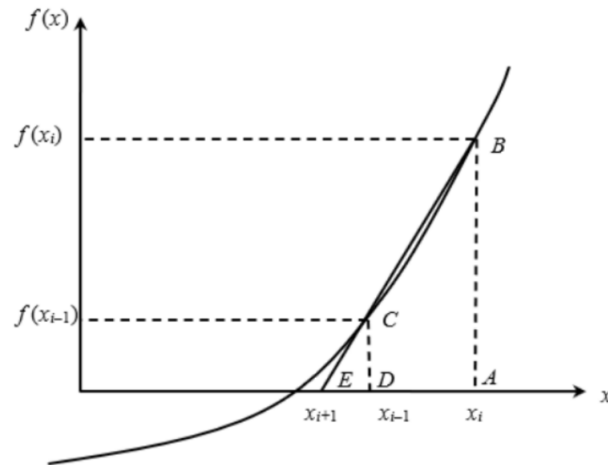


Figure 3: Geometrical representation of the Secant method.

Here, ABE and DCE are similar triangles.

Hence,

$$\begin{aligned}
 \frac{AB}{AE} &= \frac{DC}{DE} \\
 \frac{f(x_i)}{x_i - x_{i+1}} &= \frac{f(x_{i-1})}{x_{i-1} - x_{i+1}} \\
 \Rightarrow f(x_i)(x_{i-1} - x_{i+1}) &= f(x_{i-1})(x_i - x_{i+1}) \\
 \Rightarrow f(x_i)x_{i-1} - x_{i+1}f(x_i) &= x_i f(x_{i-1}) - x_{i+1}f(x_{i-1}) \\
 \Rightarrow x_{i+1}f(x_{i-1}) - x_{i+1}f(x_i) &= f(x_{i-1})x_i - f(x_i)x_{i-1} \\
 \Rightarrow x_{i+1} &= \frac{x_i f(x_{i-1}) - f(x_i)x_{i-1}}{f(x_{i-1}) - f(x_i)} + x_i - x_i \\
 \Rightarrow x_{i+1} &= x_i + \frac{x_i f(x_{i-1}) - f(x_i)x_{i-1} - x_i f(x_{i-1}) + x_i f(x_i)}{f(x_{i-1}) - f(x_i)} \\
 \Rightarrow x_{i+1} &= x_i + \frac{-f(x_i)(-x_{i-1} + x_i)}{f(x_{i-1}) - f(x_i)} \\
 \Rightarrow x_{i+1} &= x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}
 \end{aligned}$$

Hence, we get the formula for the Secant method,

$$x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

- c) The *Bronze Ratio*, often denoted by the Greek letter β (beta), is the ratio of two quantities a and b , such that $a > b > 0$, if it satisfies the equality $\frac{3a+b}{a} = \frac{a}{b}$. The constant β is equal to the positive root of a particular function $f(x)$. The numerical value of that root is,

$$x_{root}^+ = \beta = \frac{3 + \sqrt{13}}{2} = 3.302775638 \dots$$

- i. From the given scenario, formulate the function $f(x)$.

3
(CO1)
(PO2)

Solution:

According to the given condition,

$$\begin{aligned} \frac{3a+b}{a} &= \frac{a}{b} = \beta \\ \Rightarrow \frac{3a}{a} + \frac{b}{a} - \frac{a}{b} &= 0 \quad \Rightarrow \quad 3 + \frac{1}{\beta} - \beta = 0 \quad \Rightarrow \quad 3\beta + 1 - \beta^2 = 0 \\ \Rightarrow \beta^2 - 3\beta - 1 &= 0 \\ \Rightarrow x^2 - 3x - 1 &= 0 \end{aligned}$$

So, the function is $f(x) = x^2 - 3x - 1$.

- ii. Apply the Secant method to estimate the value of the bronze ratio β . The initial guesses are $x_{-1} = 1.5$ and $x_0 = 2$. Demonstrating the step-by-step mathematical procedure, conduct 4 iterations, and find the relative approximate error ($|\epsilon_a|\%$) and the number of significant digits that are at least correct (m) at the end of each iteration. Draw Table 1 on your answer script and fill it out after performing the necessary calculations.

11
(CO2)
(PO1)

Table 1: The relevant values obtained in the answer of Question 1.c)

Iteration	x_{i-1}	x_i	x_{i+1}	$ \epsilon_a \%$	m	$f(x_{i+1})$
1	1.5	2				
2						
3						
4						

Solution:

After performing the series of calculations, we end up with the following table.

Iteration	x_{i-1}	x_i	x_{i+1}	$ \epsilon_a \%$	m	$f(x_{i+1})$
1	1.5	2	8	75	0	39
2	2	8	2.428	229.41	0	-2.387
3	8	2.428	2.75	11.69	0	-1.687
4	2.428	2.75	3.524	21.98	0	0.849

2. a) “The magnitude of the true error and the magnitude of the approximate error do not necessarily reflect how bad the error is.” — Do you agree? Justify your answer.

4
(CO3)
(PO1)

Solution:

Yes, I agree with the given statement.

Let's consider an arbitrary function $f_1(x)$. Suppose we get a true error value of $E_t = p$, where p is a small number, while computing the value of $f_1'(2)$ with $h = 0.3$. Now, if we try to do the same thing for another function $f_2(x)$, where $f_2(x) = 10^{-6} \times f_1(x)$, we will get an even more minuscule true error value of $E_t = p \times 10^{-6}$. Even though the problems are the same in essence, we are getting a smaller true error value in the 2nd case, which is giving us an improper interpretability or a false impression of the goodness of the approximation. The same argument holds true for the approximate error as well.

- b) The Taylor series for a function $f(x)$ is —

15
(CO2)
(PO1)

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + f^{(5)}(x)\frac{h^5}{5!} + \dots$$

Using the Taylor series (with at least the first 5 terms), derive the Maclaurin series of

$$f(x) = \operatorname{cosec}(x)$$

Solution:

The Maclaurin series of a transcendental function is simply a Taylor series around the point $x = 0$.

We know, $\frac{d}{dx}(\operatorname{cosec}(x)) = -\operatorname{cosec}(x) \cot(x)$, and $\operatorname{cosec}(0) = \text{undefined}$. So we will get undefined values for $f(0)$ and $f^i(0)$.

Instead let's consider, $f(x) = \operatorname{cosec}(x) = \frac{1}{\sin(x)}$, and find the Maclaurin series of the denominator $g(x) = \sin(x)$. Let's calculate its derivatives at the point $x = 0$.

$$\begin{aligned} g(x) &= \sin(x), & g(0) &= 0 \\ g'(x) &= \cos(x), & g'(0) &= 1 \\ g''(x) &= -\sin(x), & g''(0) &= 0 \\ g'''(x) &= -\cos(x), & g'''(0) &= -1 \\ g^{(4)}(x) &= \sin(x), & g^{(4)}(0) &= 0 \end{aligned}$$

Using the first 5 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + \dots$$

For $x = 0$,

$$g(0+h) = g(0) + g'(0)\frac{h}{1!} + g''(0)\frac{h^2}{2!} + g'''(0)\frac{h^3}{3!} + g^{(4)}(0)\frac{h^4}{4!} + \dots$$

$$\Rightarrow g(h) = 0 + \frac{h}{1!} + 0 - \frac{h^3}{3!} + 0 + \dots$$

$$\Rightarrow g(x) = \frac{x}{1!} - \frac{x^3}{3!} + \dots \Rightarrow \sin(x) = \frac{x}{1!} - \frac{x^3}{3!} + \dots = x - \frac{x^3}{6} + \dots$$

Now, for the original function $f(x) = \operatorname{cosec}(x)$,

$$\operatorname{cosec}(x) = \frac{1}{x - \frac{x^3}{6} + \dots}$$

Performing algebraic long division, we get,

$$\begin{array}{r}
 x - \frac{x^3}{6} + \dots \overline{) \frac{\frac{1}{x} + \frac{x}{6} + \dots}{1 + 0}} \\
 \underline{1 - \frac{x^2}{6} + \dots} \\
 \frac{x^2}{6} + \dots \\
 \underline{\frac{x^2}{6} + \dots} \\
 0 + \dots \\
 \vdots
 \end{array}$$

Hence, we can write,

$$\Rightarrow \operatorname{cosec}(x) = \frac{1}{x} + \frac{x}{6} + \dots \quad [\text{Quotient of the above algebraic long division.}]$$

- c) We know, $\sin\left(\frac{\pi}{2}\right) = 1$ and $\cos\left(\frac{\pi}{2}\right) = 0$. Using this information and the Taylor series (with at least the first 5 terms) approximate the value of $\sin(2)$.

8

(CO2)

Note: The unit of angular measure that has been used here is radians (rad).

(PO1)

Solution:

Let, $x = \frac{\pi}{2}$ and $x + h = 2$. This implies, $h = 2 - x = 2 - \frac{\pi}{2} = 0.4292$.

Let's calculate its derivatives at the point $x = \frac{\pi}{2}$.

$$f(x) = \sin(x), \quad f\left(\frac{\pi}{2}\right) = 1$$

$$f'(x) = \cos(x), \quad f'\left(\frac{\pi}{2}\right) = 0$$

$$f''(x) = -\sin(x), \quad f''\left(\frac{\pi}{2}\right) = -1$$

$$f'''(x) = -\cos(x), \quad f'''\left(\frac{\pi}{2}\right) = 0$$

$$f^{(4)}(x) = \sin(x), \quad f^{(4)}\left(\frac{\pi}{2}\right) = 1$$

Using the first 5 terms of the Taylor series,

$$f(x + h) = f(x) + f'(x) \frac{h}{1!} + f''(x) \frac{h^2}{2!} + f'''(x) \frac{h^3}{3!} + f^{(4)}(x) \frac{h^4}{4!} + \dots$$

For $x = \frac{\pi}{2}$,

$$f\left(\frac{\pi}{2} + h\right) = f\left(\frac{\pi}{2}\right) + f'\left(\frac{\pi}{2}\right) \frac{h}{1!} + f''\left(\frac{\pi}{2}\right) \frac{h^2}{2!} + f'''\left(\frac{\pi}{2}\right) \frac{h^3}{3!} + f^{(4)}\left(\frac{\pi}{2}\right) \frac{h^4}{4!} + \dots$$

$$\Rightarrow f\left(\frac{\pi}{2} + 0.4292\right) = 1 + 0 - \frac{(0.4292)^2}{2!} + 0 + \frac{(0.4292)^4}{4!} + \dots$$

$$= 1 + 0 - 0.092106 + 0 + 0.00141393 + \dots$$

$$\approx 0.90931$$

3. a) Prove that a polynomial of degree n or less that passes through $n + 1$ data points is unique.

6
(CO4)
(PO1)

Solution:

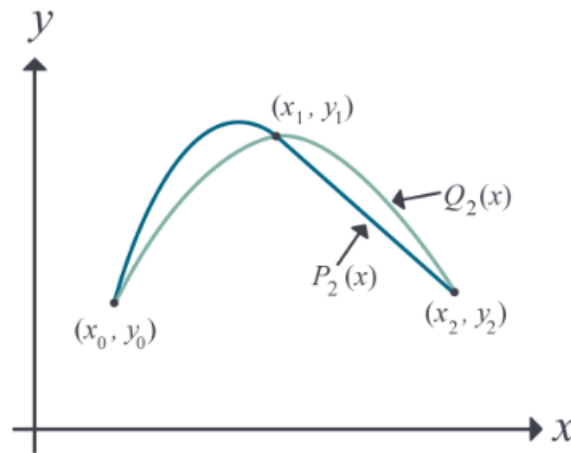


Figure 4: Interpolating polynomials through three points.

Proof: Let us use proof by contradiction. If the polynomial is not unique, then at least two polynomials of order n or less pass through the $n + 1$ data points.

Assume two polynomials $P_n(x)$ and $Q_n(x)$ go through $n + 1$ data points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$. Then,

$$R_n(x) = P_n(x) - Q_n(x)$$

Since $P_n(x)$ and $Q_n(x)$ pass through all the $n + 1$ data points, we can say,

$$P_n(x_i) = Q_n(x_i) \quad \forall i \in \{0, \dots, n\}$$

Hence

$$\begin{aligned} R_n(x_i) &= P_n(x_i) - Q_n(x_i) \quad \forall i \in \{0, \dots, n\} \\ \Rightarrow R_n(x_i) &= 0 \end{aligned}$$

The n th order polynomial $R_n(x)$ has $n + 1$ zeros. A polynomial of order n can have $n + 1$ zeros only if it is identical to a zero polynomial, that is,

$$\begin{aligned} R_n(x) &\equiv 0 \\ \Rightarrow P_n(x) &\equiv Q_n(x) \end{aligned}$$

So, $P_n(x)$ and $Q_n(x)$ must be the same unique polynomial, which contradicts our initial assumption. $\square$ Q.E.D.

- b) Compare and contrast the following methods of interpolation — Direct method, Newton's Divided Difference method, and Lagrangian method.

4
(CO3)
(PO1)

Solution:

Any 2 valid comparisons will suffice. Some comparisons can be—

Criteria	Direct	Newton's Divided Difference	Lagrangian
Interpolant Polynomial Representation	$f_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$	$f_n(x) = b_0 + b_1(x - x_0) + \dots + b_n(x - x_0)(x - x_1) \dots (x - x_{n-1})$	$f_n(x) = \sum_{i=0}^n \left(\prod_{j=0, j \neq i}^n \frac{x - x_j}{x_i - x_j} \right) f(x_i)$
Computational Complexity	Depends on the algorithm used to solve the equations. It may be $O(n^3)$, $O(n^3 \log(\ A\ + \ b\))$, or $O((n+1)!)$.	Using memoized dynamic programming, we get a complexity of $O(n^2)$.	The naïve implementation is $O(n^2)$, but it can be optimized further down to $O(n \log^2(n))$.
Property of the terms	Each term has a coefficient a_i and these coefficients are obtained by solving all the equations simultaneously.	Each term has a coefficient b_i and each of these coefficients can be obtained by solving one equation.	The polynomial is a weighted sum of the given functional values $f(x_i)$, and each term is a product of the Lagrangian weight function $L_i(x)$ and the functional value $f(x_i)$.
Vulnerable to the Runge Phenomenon?	Yes	Yes	Yes

- c) A thermistor, also known as thermal resistor, is a semiconductor type of resistor whose resistance is strongly dependent on temperature. To measure temperature using a thermistor, the manufacturers provide users with a Temperature (T) vs. Resistance (R) calibration curve. If we measure the resistance of the thermistor using an Ohmmeter, then we can refer to the aforementioned curve and determine the corresponding temperature value. Table 2 portrays multiple recorded observations involving a thermistor.

Table 2: Temperature T of a thermistor as a function of its resistance R for Question-3.c)

Resistance, R (Ω)	Temperature, T ($^{\circ}C$)
1101	25.113
911.3	30.131
636	40.12
451.1	50.128

Determine the value of the temperature T when the thermistor resistance is $R = 754.8\Omega$ using the Newton's Divided Difference method of interpolation and a second order polynomial.

Solution:

The 2nd order NDD interpolant is —

$$T(R) = b_0 + b_1(R - R_0) + b_2(R - R_0)(R - R_1)$$

Given, $R = 754.8\Omega$, and the 3 nearest data points that bracket it are, $R_0 = 911.3$, $R_1 = 636$, and $R_2 = 451.1$. The corresponding T values are $T(R_0) = 30.131$, $T(R_1) = 40.12$, and $T(R_2) = 50.128$ respectively.

Now, let's calculate the coefficients.

$$b_0 = T(R_0) = 30.131$$

$$b_1 = \frac{T(R_1) - T(R_0)}{R_1 - R_0} = \frac{40.12 - 30.131}{636 - 911.3} = -0.0362$$

$$b_2 = \frac{\frac{T(R_2) - T(R_1)}{R_2 - R_1} - \frac{T(R_1) - T(R_0)}{R_1 - R_0}}{R_2 - R_0} = \frac{\frac{50.128 - 40.12}{451.1 - 636} - \frac{40.12 - 30.131}{636 - 911.3}}{451.1 - 911.3} = 3.877 \times 10^{-5}$$

Hence,

$$T(R) = b_0 + b_1(R - R_0) + b_2(R - R_0)(R - R_1)$$

$$= 30.131 - 0.0362(R - 911.3) + 3.877 \times 10^{-5}(R - 911.3)(R - 636); \quad 451.1 \leq R \leq 911.3$$

Now that we have the NDD interpolant, we can obtain the value of $T(754.8)$.

$$\begin{aligned} T(754.8) &= 30.131 - 0.0362(754.8 - 911.3) + 3.877 \times 10^{-5}(754.8 - 911.3)(754.8 - 636) \\ &\approx 35.075^{\circ}C \end{aligned}$$

4. a) The general linear regression model for predicting the response for a given set of n data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ is 6 + 6
(CO4)
(PO1)

$$y = a_0 + a_1x$$

where a_0 and a_1 are the two parameters of the regression model.

- i. Derive the formulae for finding the optimal values of a_0 and a_1 .

Solution:

The residual error for each given data point is,

$$E_i = y_i - a_0 - a_1x_i$$

So the sum of the square of the residual errors is,

$$S_r = \sum_{i=1}^n E_i^2 = \sum_{i=1}^n (y_i - a_0 - a_1x_i)^2$$

To find the optimal a_0 and a_1 , we look for the values of a_0 and a_1 for which S_r is the absolute minimum. At first, let's do the first derivative test. Taking the partial derivatives of S_r with respect to a_0 and a_1 and set them to zero.

$$\frac{\partial S_r}{\partial a_0} = 2 \sum_{i=1}^n (y_i - a_0 - a_1x_i)(-1) = 0$$

$$\frac{\partial S_r}{\partial a_1} = 2 \sum_{i=1}^n (y_i - a_0 - a_1x_i)(-x_i) = 0$$

Dividing both sides by 2 and expanding the summations in the aforementioned equations,

$$-\sum_{i=1}^n y_i + \sum_{i=1}^n a_0 + \sum_{i=1}^n a_1x_i = 0 \Rightarrow -\sum_i y_i + na_0 + \sum_i a_1x_i = 0 \Rightarrow a_0 = \frac{\sum_i y_i}{n} - a_1 \frac{\sum_i x_i}{n}$$

For the 2nd equation, we perform substitution

$$-\sum_{i=1}^n y_i x_i + \sum_{i=1}^n a_0 x_i + \sum_{i=1}^n a_1 x_i^2 = 0 \Rightarrow \sum_i x_i y_i - \left(\frac{\sum_i y_i}{n} - a_1 \frac{\sum_i x_i}{n} \right) \sum_i x_i - a_1 \sum_i x_i^2 = 0$$

$$\Rightarrow \sum_i x_i y_i - \sum_i x_i \frac{\sum_i y_i}{n} + a_1 \frac{(\sum_i x_i)^2}{n} - a_1 \sum_i x_i^2 = 0$$

$$\Rightarrow \left(\frac{\left(\sum_i x_i \right)^2}{n} - \sum_i x_i^2 \right) = \frac{\sum_i x_i \sum_i y_i}{n} - \sum_i x_i y_i$$

$$\Rightarrow a_1 = \frac{n \sum_i x_i y_i - \sum_i x_i \sum_i y_i}{n \sum_i x_i^2 - \left(\sum_i x_i \right)^2} = \frac{\sum_i x_i y_i - n \bar{x} \bar{y}}{\sum_i x_i^2 - n \bar{x}^2}$$

Plugging in this expression for a_1 to the previous equation,

$$a_0 = \frac{\sum_i x_i^2 \sum_i y_i - \sum_i x_i \sum_i x_i y_i}{n \sum_i x_i^2 - \left(\sum_i x_i \right)^2}$$

- ii. Prove that the values of a_0 and a_1 obtained using the formulae derived in Question-4.a)(i) correspond to the absolute minimum of the optimization criterion used.

Solution:

Let's invoke the second derivative for this.

From the 1st-order partial derivatives in the last question, we get,

$$\frac{\partial^2 S_r}{\partial a_0^2} = -2 \sum_i -1 = 2n > 0; \quad [2\text{nd condition satisfied.}]$$

$$\frac{\partial^2 S_r}{\partial a_1^2} = 2 \sum_i x_i^2 > 0; \quad [2\text{nd condition satisfied.}]$$

$$\frac{\partial^2 S_r}{\partial a_0 \partial a_1} = 2 \sum_i x_i$$

Now, let's test the 1st condition of the second derivative test.

$$\begin{aligned} \frac{\partial^2 S_r}{\partial a_0^2} \frac{\partial^2 S_r}{\partial a_1^2} - \left(\frac{\partial^2 S_r}{\partial a_0 \partial a_1} \right)^2 &= (2n) \left(2 \sum_i x_i^2 \right) - \left(2 \sum_i x_i \right)^2 \\ &= 4 \left[n \sum_i x_i^2 - \left(\sum_i x_i \right)^2 \right] > 0; \quad [1\text{st condition satisfied.}] \end{aligned}$$

So, the critical point (a_0, a_1) obtained using the formulae corresponds to the absolute minimum value of S_r .

- b) Suppose you are given a set of n data points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ and you want to regress the data to an m th order Polynomial model, where $m < n$. How would you choose a suitable value of m ? 5
(CO3)
(PO1)

Solution:

In order to choose a suitable value of m , we need to maintain a Bias-Variance tradeoff. If the value of m is too high, the polynomial model will overfit the given n data points and result in high variance. On the other hand, if the value of m is too small, the polynomial model will underfit the given n data points and result in high bias.

- c) Sodium borohydride (NaBH_4) is an inorganic compound that can be used as fuel for fuel cells. The Overpotential (η) vs Current (I) data depicted in Table 3 was obtained during a study conducted to assess its electrochemical kinetics. 8
(CO2)
(PO1)

Table 3: Electrochemical Kinetics data of NaBH_4 for Question-4.c)

Overpotential, η (V)	Current, I (A)
-0.29563	0.00226
-0.24346	0.00212
-0.19012	0.00206
-0.18772	0.00202
-0.13407	0.00199
-0.0861	0.00195

The relationship that exists between the overpotential (η) and current (I) can be represented

using the Logarithmic model (also known as the Tafel equation) —

$$\eta = a + b \ln I$$

where a is an electrochemical kinetics parameter of borohydride on the electrode and b is the experimental Tafel slope parameter. Determine the value of a and b using the data points provided in Table 3. You are allowed to apply data transformation/linearization.

Solution:

Suppose, $x = \ln I$ and $y = \eta$. Then we can express the Tafel equation as a straight line $y = a + bx$. We can now obtain the value of the parameters a and b using the linear regression formulae,

$$b = \frac{n \sum_i x_i y_i - \sum_i x_i \sum_i y_i}{n \sum_i x_i^2 - \left(\sum_i x_i \right)^2}$$

$$a = \bar{y} - b\bar{x}$$

Using the $\ln I$ and η values in the calculator, we find the following values,
 $\sum_i x_i = -37.0979$, $\sum_i y_i = -1.1371$, $\sum_i x_i^2 = 229.3896$, $\sum_i x_i y_i = 7.0116$.

Let's plug these values into the formulae.

$$b = \frac{6 \times 7.0116 - (-37.0979) \times (-1.1371)}{6 \times 229.3896 - (-37.0979)^2} = -1.3601$$

$$a = \frac{-1.1371}{6} - (-1.3601) \times \frac{-37.0979}{6} = -8.5992$$

Plugging these values into the given Logarithmic model equation, we get,

$$\eta = -8.5992 - 1.3601 \times \ln I$$

5. a) i. Using the method of coefficients, derive the formula for the Single-segment Trapezoidal rule.

7 + 6
(CO3)
(PO1)

Solution:

Let's consider the integral to be a weighted sum of the functional values at the two limits a and b .

$$\int_a^b f(x) dx \approx c_1 f(a) + c_2 f(b)$$

This weighted sum is exactly equal to the integral of the straight line $f_1(x) = a_0 + a_1 x$.

$$\int_a^b (a_0 + a_1 x) dx = \left[a_0 x + a_1 \frac{x^2}{2} \right]_a^b = a_0(b - a) + a_1 \left(\frac{b^2 - a^2}{2} \right)$$

We know that the straight line passes through the points $(a, f(a))$ and $(b, f(b))$. So we can express the functional values as,

$$f(a) = a_0 + a_1 a$$

$$f(b) = a_0 + a_1 b$$

Then we can write the weighted sum as,

$$c_1(a_0 + a_1 a) + c_2(a_0 + a_1 b) = c_1 a_0 + c_1 a_1 a + c_2 a_0 + c_2 a_1 b$$

$$= a_0(c_1 + c_2) + a_1(c_1 a + c_2 b)$$

Equating the coefficients, we get,

$$\begin{aligned} c_1 + c_2 &= b - a \\ c_1 a + c_2 b &= \frac{b^2 - a^2}{2} \end{aligned}$$

Solving them, we get,

$$c_1 = c_2 = \frac{b - a}{2}$$

Therefore,

$$\begin{aligned} \int_a^b f(x) dx &\approx c_1 f(a) + c_2 f(b) \\ &= \frac{b - a}{2} f(a) + \frac{b - a}{2} f(b) \end{aligned}$$

- ii. Applying the formula from Question-5.a)(i), derive the formula for the Multi-segment Trapezoidal rule.

Solution:

Taking a step size of $h = \frac{b - a}{n}$, we break down the integral I into n integrals in the following manner,

$$\begin{aligned} I &= \int_a^b f(x) dx \\ &= \int_a^{a+h} f(x) dx + \int_{a+h}^{a+2h} f(x) dx + \cdots + \int_{a+(n-2)h}^{a+(n-1)h} f(x) dx + \int_{a+(n-1)h}^{a+nh=b} f(x) dx \\ &= [(a + h) - a] \left[\frac{f(a) + f(a + h)}{2} \right] + \cdots + [b - (a + (n - 1)h)] \left[\frac{f(a + (n - 1)h) + f(b)}{2} \right] \\ &= h \left[\frac{f(a) + 2f(a + h) + \cdots + 2f(a + (n - 1)h) + f(b)}{2} \right] \\ &= \frac{h}{2} \left[f(a) + 2 \left\{ \sum_{i=1}^{n-1} f(a + ih) \right\} + f(b) \right] \\ &= \frac{b - a}{2n} \left[f(a) + 2 \left\{ \sum_{i=1}^{n-1} f(a + ih) \right\} + f(b) \right] \end{aligned}$$

b) The velocity v of a body is given by

$$v(t) = \begin{cases} 2t; & \text{if } 1 \leq t \leq 5 \\ 5t^2 + 3; & \text{if } 5 < t \leq 14 \end{cases}$$

where the time t is given in seconds (s), and v is given in meters per second (m s^{-1}).

- i. Using 2-segment Simpson's 1/3 rule, approximate the distance (S) in meters covered by the body from $t = 2\text{s}$ to $t = 9\text{s}$.

Solution:

The multiple-segment equation for Simpson's 1/3 rule is

$$\int_a^b f(x)dx \approx \frac{b-a}{3n} \left[f(x_0) + 4 \sum_{\substack{i=1, \\ i=\text{odd}}}^{n-1} f(x_i) + 2 \sum_{\substack{i=2, \\ i=\text{even}}}^{n-2} f(x_i) + f(x_n) \right]$$

We have, $a = 2$, $b = 9$, $n = 2$, $h = \frac{b-a}{n} = \frac{9-2}{2} = 3.5$. We also know that the distance function $S(t)$ is the integration of the velocity function $v(t)$. The distance covered by the body is, therefore,

$$\begin{aligned} S &= \int_2^9 v(t)dt \approx \frac{9-2}{3 \times 2} \left[v(t_0) + 4 \sum_{\substack{i=1, \\ i=\text{odd}}}^{2-1} v(t_i) + 2 \sum_{\substack{i=2, \\ i=\text{even}}}^{2-2} v(t_i) + v(t_n) \right] \\ &= \frac{9-2}{3 \times 2} \times [v(t_0) + 4v(t_1) + v(t_2)] \\ &= 1.1667 \times [2 \times 2 + 4 \times 5 \times 5.5^2 + 3 + 5 \times 9^2 + 3] \\ &= 1.1667 \times [4 + 4 \times 154.25 + 408] \\ &= 1200.5 \text{ m} \end{aligned}$$

- ii. Analytically determine the exact value of the covered distance (S_{exact}) and use it to calculate the true error (E_t) and the absolute relative true error ($|\epsilon_t|\%$).

Solution:

The exact value of the covered distance is,

$$\begin{aligned} S_{\text{exact}} &= \int_2^5 2t dt + \int_5^9 (5t^2 + 3) dt \\ &= \left[\frac{2t^2}{2} \right]_2^5 + \left[\frac{5t^3}{3} + 3t \right]_5^9 \\ &= 1039.6667 \text{ m} \end{aligned}$$

The true error is,

$$E_t = \text{True Value} - \text{Approx. Value} = 1039.6667 \text{ m} - 1200.5 \text{ m} = -160.8333 \text{ m}$$

The absolute relative true error is,

$$|\epsilon_t| = \left| \frac{\text{True Value} - \text{Approx. Value}}{\text{True Value}} \right| \times 100\% = 15.469\%$$

c) Define Eigenvector and Eigenvalue.

3
(CO3)
(PO1)

Solution:

The vectors $\vec{v}$ of a vector space that do not get knocked off their own spans upon undergoing linear transformation by a matrix A , but are just stretched or squished by a certain factor λ , are called the eigenvectors of the linear transformation done by A .

Mathematically,

$$A\vec{v} = \lambda\vec{v}$$

Here, the corresponding eigenvalue is the factor by which an eigenvector is stretched or squished. If the eigenvalue is negative, the eigenvector's direction is reversed.

6. a) Compare and contrast the following numerical algorithms for solving 1st-order ordinary differential equations — Euler's method, Runge-Kutta 2nd order method, and Runge-Kutta 4th order method.

4
(CO3)
(PO1)

Solution:

Any 2 valid comparisons will suffice. Some comparisons can be—

Criteria	Euler	Runge-Kutta 2nd Order	Runge-Kutta 4th Order
Formula	$y_{i+1} = y_i + f(x_i, y_i)h$	$y_{i+1} = y_i + (a_1k_1 + a_2k_2)h, k_1 = f(x_i, y_i), k_2 = f(x_i + ph, y_i + qk_1h)$	$y_{i+1} = y_i + (a_1k_1 + a_2k_2 + a_3k_3 + a_4k_4)h$
Terms of the Taylor series	First 2 terms.	First 3 terms.	First 5 terms.
Order of the Taylor polynomial	1st order — which is why it's also called the Runge-Kutta 1st order method.	2nd order — which is why it's called the Runge-Kutta 2nd order method.	4th order — which is why it's also called the Runge-Kutta 4th order method.
Local Truncation Error	$O(h^2)$	$O(h^3)$	$O(h^5)$
Global Truncation Error	$O(h)$	$O(h^2)$	$O(h^4)$
Variant Approaches	None.	Has 3 different versions – Heun's method, Midpoint method, and Ralston's method	Has 2 different versions – Runge's approach and Kutta's approach.

b) Determine the approximate value of

$$I = \int_5^8 6x^3 dx$$

using Euler's method. Use a step size of $h = 1.5$.

Solution:

Let's rewrite the given integral as the solution of an ODE –

$$\frac{dy}{dx} = 6x^3; \quad y(5) = 0$$

where $y(8)$ gives us the value of the integral.

The Euler's method formula is,

$$y_{i+1} = y_i + f(x_i, y_i)h$$

In the 1st step, we have $x_0 = 5$, $y_0 = 0$, and given $h = 1.5$.

$$x_1 = x_0 + h = 5 + 1.5 = 6.5$$

$$\begin{aligned} y_1 &= y_0 + f(x_0, y_0)h = 0 + f(5, 0) \times 1.5 \\ &= 0 + (6 \times 5^3) \times 1.5 = 1125 \approx y(6.5) \end{aligned}$$

In the 2nd step, we have $x_1 = 6.5$, $y_1 = 1125$, and as before $h = 1.5$.

$$x_2 = x_1 + h = 6.5 + 1.5 = 8$$

$$\begin{aligned} y_2 &= y_1 + f(x_1, y_1)h = 1125 + f(6.5, 1125) \times 1.5 \\ &= 1125 + 6 \times 6.5^3 \times 1.5 = 3596.625 \approx y(8) \end{aligned}$$

Hence,

$$\int_5^8 6x^3 dx = y(8) - y(5) \approx 3596.625 - 0 = 3596.625$$

c) i. Determine the *minimum* number of significant digits in each of these numerical values

(a) 069420

(b) 6.94200×10^5

(c) 0.069420

(d) 69.00420

(e) 69.0

(f) 69420

3 + 2
(CO4)
(PO1)

Solution:

We need to determine the *minimum* number of significant digits.

- (a) 069420 has 4 significant digits, since leading 0's are not significant and trailing 0's in an integer may or may not be significant.
- (b) 6.94200×10^5 has 6 significant digits, since all digits of the significand/coefficient of a number's representation using Scientific Notation are significant.
- (c) 0.069420 has 5 significant digits, since leading 0's are not significant and trailing 0's in a number with the decimal point are significant.
- (d) 69.00420 has 7 significant digits, since 0's between two significant non-zero digits are significant and trailing 0's in a number with the decimal point are significant.
- (e) 69.0 has 3 significant digits, since 0's to the right of the decimal point are significant.
- (f) 69420 has 4 significant digits, since trailing 0's in an integer may or may not be significant.

ii. Determine the order and degree of each of these higher-order differential equations

(a) $x \frac{dy}{dx} - \left(\frac{d^2y}{dx^2} \right)^3 + 5 = 0$

(b) $\left(\frac{dy}{dx} \right)^2 - 4xy = 3$

(c) $\frac{d^3y}{dx^3} + x^4 \frac{d^2y}{dx^2} \left(\frac{dy}{dx} \right)^4 = \sin x$

(d) $\frac{dy}{dx} + xy = x^2$

Solution:

The orders and degrees of these higher-order ODEs are,

(a) Order = 2, Degree = 3.

(b) Order = 1, Degree = 2.

(c) Order = 3, Degree = 1.

(d) Order = 1, Degree = 1.

d) i. Determine the LU Decomposition of the 3×3 matrix

5 + 5

(CO2)

(PO1)

$$A = \begin{bmatrix} 6 & 18 & 3 \\ 2 & 12 & 1 \\ 4 & 15 & 3 \end{bmatrix}$$

Solution:

The LU Decomposition can be done as follows,

$$\begin{bmatrix} 6 & 18 & 3 \\ 2 & 12 & 1 \\ 4 & 15 & 3 \end{bmatrix} \xrightarrow[E_{21}]{R_2 = -\frac{R_1}{3} + R_2} \begin{bmatrix} 6 & 18 & 3 \\ 0 & 6 & 0 \\ 4 & 15 & 3 \end{bmatrix} \xrightarrow[E_{31}]{R_3 = -\frac{2R_1}{3} + R_3} \begin{bmatrix} 6 & 18 & 3 \\ 0 & 6 & 0 \\ 0 & 3 & 1 \end{bmatrix} \xrightarrow[E_{32}]{R_3 = -\frac{R_2}{2} + R_3} \begin{bmatrix} 6 & 18 & 3 \\ 0 & 6 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

where,

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{3} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -\frac{2}{3} & 0 & 1 \end{bmatrix} \quad E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{1}{2} & 1 \end{bmatrix}$$

So we have,

$$L = \begin{bmatrix} 1 & 0 & 0 \\ \frac{1}{3} & 1 & 0 \\ \frac{2}{3} & \frac{1}{2} & 1 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 6 & 18 & 3 \\ 0 & 6 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- ii. Suppose $u = u(x, y)$ is a function of two variables that we only know at discrete grid points (x_i, y_j) given in the matrix

$$[u_{i,j}] = \begin{bmatrix} 5.1 & 6.5 & 7.5 & 8.1 & 8.4 \\ 5.5 & 6.8 & 7.8 & 8.3 & 8.9 \\ 5.5 & 6.9 & 9.0 & 8.4 & 9.1 \\ 5.4 & 9.6 & 9.1 & 8.6 & 9.4 \end{bmatrix}$$

Find the approximate value of the following partial derivatives

- (a) $u_x(x_2, y_4)$ (b) $u_y(x_2, y_4)$ (c) $u_{xx}(x_2, y_4)$
 (d) $u_{yy}(x_2, y_4)$ (e) $u_{xy}(x_2, y_4)$

using the central difference formula. Consider $h = 0.5$ and $k = 0.2$.

Note: Assume 1-based indexing.

Solution:

$$(a) \quad u_x(x_2, y_4) \approx \frac{u_{3,4} - u_{1,4}}{2h} = \frac{8.4 - 8.1}{2 \times 0.5} = 0.3.$$

$$(b) \quad u_y(x_2, y_4) \approx \frac{u_{2,5} - u_{2,3}}{2k} = \frac{8.9 - 7.8}{2 \times 0.2} = 2.75.$$

$$(c) \quad u_{xx}(x_2, y_4) \approx \frac{u_{3,4} - 2u_{2,4} + u_{1,4}}{h^2} = \frac{8.4 - 2 \times 8.3 + 8.1}{0.5^2} = -0.4.$$

$$(d) \quad u_{yy}(x_2, y_4) \approx \frac{u_{2,5} - 2u_{2,4} + u_{2,3}}{k^2} = \frac{8.9 - 2 \times 8.3 + 7.8}{0.2^2} = 2.5.$$

$$(e) \quad u_{xy}(x_2, y_4) \approx \frac{u_{3,5} - u_{3,3} - u_{1,5} + u_{1,3}}{4hk} = \frac{9.1 - 9.0 - 8.4 + 7.5}{4 \times 0.5 \times 0.2} = -2.$$